

1. Administrative & Study Identification

Full Study Title: A Randomized, Participant- and Investigator-blinded, Controlled, Parallel Group Study to Assess the Efficacy, Safety, Tolerability, Pharmacokinetics and Pharmacodynamics of LXE408 in Participants With Chronic Chagas Disease Without Severe Organ Dysfunction **Short Title/Acronym:** A Study of Efficacy, Safety, Tolerability of LXE408 in Participants With Chronic Chagas Disease. **Protocol Number:** CLXE408B12201 **NCT Number:** NCT06632600 **Posting ID:** 929550737447734735 **Trial Status:** Recruiting **Sponsor Name:** Novartis **Investigational Product:** LXE408 (kinetoplastid-selective proteasome inhibitor) **Phase of Development:** Phase 2 (Proof of Concept) **Medical Monitor:** Novartis Pharmaceuticals Clinical Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the ability of LXE408 to clear or reduce the level of *Trypanosoma cruzi* parasites in the blood of patients with chronic Chagas disease (anti-parasitological activity) compared to placebo and the standard of care, benznidazole. **Secondary Objectives:** To assess the safety, tolerability, pharmacokinetics (PK), and pharmacodynamics (PD) of LXE408.

2.2 Background & Rationale To address the need for a shorter, highly effective, and safer oral treatment for Chronic Chagas disease. The current standard treatment (benznidazole) requires a full 60-day course and is often associated with adverse effects. LXE408 targets the parasite in a novel way and is being tested in shorter regimens (14 to 28 days) to potentially improve patient outcomes and medication adherence while minimizing toxicity.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled and Active Comparator (Benznidazole) **Blinding:** Double-blind (Participant- and Investigator-blinded) **Randomization:** Randomized (Parallel Group, 4 treatment arms)

3.2 Planned Enrollment & Duration Target \$N\$: 130 participants **Number of Sites:** Global/Multicenter **Estimated Study Start:** 28-Apr-2025 **Estimated Primary Completion:** 22-Mar-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female participants aged ≥ 18 years to ≤ 60 years old. 2. Confirmed diagnosis of *T. cruzi* infection. 3. History that the participant has been determined to be in the chronic phase of Chagas disease. 4. Considered by the investigator eligible for and able to comply with local prescribing information for benznidazole. 5. Ability and willingness to communicate well with the investigator/study site and comply with study requirements, with written informed consent obtained.

4.2 Exclusion Criteria 1. Signs and/or symptoms of Chagas disease in the acute phase at screening. 2. History of Chagas disease treatment with benznidazole or nifurtimox at any time in the past. 3. Known or suspected ongoing, chronic, or recurrent viral, bacterial, or fungal infectious diseases (e.g., Tuberculosis, leishmaniasis, severe malaria, atypical mycobacterial infection, listeriosis, aspergillosis, endemic mycoses, or documented HIV infection). 4. History of and/or current symptoms or signs of moderate or severe CD-related gastrointestinal disease, or documented clinically significant cardiovascular conditions (e.g., unstable ischemic heart disease, NYHA Class III/IV heart failure, arrhythmias). 5. Participants who weigh < 50 kg or > 90 kg at screening. 6. Severe organ dysfunction including pancreatic injury/pancreatitis (amylase/lipase $> 1.5 \times$ ULN at screening, or history of recurrent pancreatitis), abnormal liver function tests (ALT, AST, GGT, alkaline phosphatase $> 1.5 \times$ ULN), or renal disease (creatinine $> 1.5 \times$ ULN, calculated eGFR < 60 mL/min). 7. History of lymphoproliferative disease or any known malignancy within the past 5 years (with specific minor exceptions). 8. Hematology parameters outside of defined thresholds at screening. 9. Current use of prohibited medications/investigational drugs, or multiple/recurring allergies (including allergy to the investigational compound or benznidazole). 10. History of drug abuse or unhealthy alcohol use within the 12 months prior to dosing. 11. Pregnant or nursing (lactating) women, or women of child-bearing potential not using highly effective contraception. 12. Contraindications to MRI imaging (excludes participant from MRI assessments only).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: LXE408, benznidazole, or placebo administered in 4 parallel treatment arms. **Route of Administration:** Oral **Duration of Treatment:** 14 to 28 days for LXE408/Placebo; 60 days for Benznidazole. Followed by observation up to 12 months.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for non-excluded co-morbidities. **Prohibited:** Use of benznidazole in the blinded arms is prohibited

until unblinding occurs after all participants complete the Month 12 evaluation.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	2 to 4 weeks	Informed Consent, Medical History, Organ Function Labs, Baseline PCR
Baseline Day 0		Randomization, First Dose
Interim	Months 2, 4, and 6	Safety Labs, PK/PD Assessments, Parasitological evaluation (PCR)
End of Study	Month 12	Final Clinical Assessments, Exit Interview, IP Accountability, Unblinding

(Note: Multiple visits occur during the 14-60 day active dosing period for monitoring)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint - Presence or absence of sustained parasitological clearance using polymerase chain reaction (PCR) results over 6 months - LXE408 28 days versus placebo: The parasitological load will be measured using PCR at 2 months, 4 months, and 6 months. Multiple samples are evaluated at each timepoint. A participant is considered negative at a visit if the parasite is not detectable in all samples, and positive if detectable in at least one sample. Sustained clearance is achieved if the participant is negative at all 3 visits.

7.2 Secondary & Exploratory Endpoints - Presence or absence of sustained parasitological clearance using PCR results over 6 months - LXE408 14 days versus placebo. - Presence or absence of sustained parasitological clearance using PCR results over 6 months - LXE408 28 days and 14 days versus benznidazole. - Presence or absence of parasitological clearance overall. - Safety and Tolerability (monitoring of Adverse Events and Serious Adverse Events). - Pharmacokinetics (PK) and Pharmacodynamics (PD) of LXE408.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via physical examinations, vital signs, ECG, and clinical laboratory evaluations (including LFTs and amylase/lipase). **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety monitoring mechanisms managed by Novartis Pharmaceuticals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Target $N=130$ appropriately powered for a Phase 2 Proof of Concept study comparing multiple parallel arms. **Handling of Missing Data:** Standard handling for PCR dropouts utilizing pre-specified missing data imputation algorithms.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits ensuring adherence to protocol and blinding rules. **Audit Trail:** Robust electronic Case Report Form (eCRF) locking and tracking for central laboratory PCR results and data clarification forms.

11. Study Identifiers & Governance

Primary ID: CLXE408B12201 **Registry Number:** NCT06632600 **Posting ID:** 929550737447734735 **Sponsor/Lead Organization:** Novartis **Study Title:** A Study of Efficacy, Safety, Tolerability of LXE408 in Participants With Chronic Chagas Disease. **Official Regulatory Title:** A Randomized, Participant- and Investigator-blinded, Controlled, Parallel Group Study to Assess the Efficacy, Safety, Tolerability, Pharmacokinetics and Pharmacodynamics of LXE408 in Participants With Chronic Chagas Disease Without Severe Organ Dysfunction.

12. Clinical Framework (Disease Specific)

Primary Condition: {{Chagas Disease / Chronic Chagas Disease (CICD)}} **Diagnosis Threshold:** {{Confirmed positive serology/PCR for T. cruzi infection}} **Medical Methodology:** {{Trypanocidal Therapy (LXE408 vs. Benznidazole)}} **Optimization Target:** {{Sustained Parasitological Clearance in blood}}

13. Study Procedures & Methodology

Management Pathway: {{Standard clinical pathway for Chronic Chagas Disease}} **Education Protocol (HCP):** {{Investigator training for double-blind protocol adherence and PCR sampling}} **Education Protocol (Patient):** {{Counseling on strict oral medication adherence and 12-month follow-up schedule}} **Quality Audit Mechanism:** {{Central laboratory PCR analysis tracking & rigorous PK/PD monitoring}}

14. Observation & Follow-Up Schedule

Total Duration: {{12 months of follow-up}} **Onsite Visit Cadence:** {{Screening, Baseline, and evaluations at Months 2, 4, 6, and 12}} **Remote Monitoring Frequency:** {{As per site monitoring plan}} **Checklist Review Interval:** {{Ongoing throughout the 14-60 day active treatment phase}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					